

K Means Clustering with Python

Method Used

K Means Clustering is an unsupervised learning algorithm that tries to cluster data based on their similarity. Unsupervised learning means that there is no outcome to be predicted, and the algorithm just tries to find patterns in the data. In k means clustering, we have to specify the number of clusters we want the data to be grouped into. The algorithm randomly assigns each observation to a cluster, and finds the centroid of each cluster. Then, the algorithm iterates through two steps: Reassign data points to the cluster whose centroid is closest. Calculate new centroid of each cluster. These two steps are repeated till the within cluster variation cannot be reduced any further. The within cluster variation is calculated as the sum of the euclidean distance between the data points and their respective cluster centroids.

```
In [1]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

```
In [3]: data_iris = pd.read_csv('iris.csv')
```

```
In [4]: data_iris.head(60)
```

Out[4]:

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
0	1	5.1	3.5	1.4	0.2	Iris-setosa
1	2	4.9	3.0	1.4	0.2	Iris-setosa
2	3	4.7	3.2	1.3	0.2	Iris-setosa
3	4	4.6	3.1	1.5	0.2	Iris-setosa
4	5	5.0	3.6	1.4	0.2	Iris-setosa
5	6	5.4	3.9	1.7	0.4	Iris-setosa
6	7	4.6	3.4	1.4	0.3	Iris-setosa
7	8	5.0	3.4	1.5	0.2	Iris-setosa
8	9	4.4	2.9	1.4	0.2	Iris-setosa
9	10	4.9	3.1	1.5	0.1	Iris-setosa
10	11	5.4	3.7	1.5	0.2	Iris-setosa
11	12	4.8	3.4	1.6	0.2	Iris-setosa
12	13	4.8	3.0	1.4	0.1	Iris-setosa
13	14	4.3	3.0	1.1	0.1	Iris-setosa
14	15	5.8	4.0	1.2	0.2	Iris-setosa
15	16	5.7	4.4	1.5	0.4	Iris-setosa
16	17	5.4	3.9	1.3	0.4	Iris-setosa
17	18	5.1	3.5	1.4	0.3	Iris-setosa
18	19	5.7	3.8	1.7	0.3	Iris-setosa
19	20	5.1	3.8	1.5	0.3	Iris-setosa
20	21	5.4	3.4	1.7	0.2	Iris-setosa
21	22	5.1	3.7	1.5	0.4	Iris-setosa
22	23	4.6	3.6	1.0	0.2	Iris-setosa
23	24	5.1	3.3	1.7	0.5	Iris-setosa
24	25	4.8	3.4	1.9	0.2	Iris-setosa
25	26	5.0	3.0	1.6	0.2	Iris-setosa
26	27	5.0	3.4	1.6	0.4	Iris-setosa
27	28	5.2	3.5	1.5	0.2	Iris-setosa
28	29	5.2	3.4	1.4	0.2	Iris-setosa
29	30	4.7	3.2	1.6	0.2	Iris-setosa
30	31	4.8	3.1	1.6	0.2	Iris-setosa
31	32	5.4	3.4	1.5	0.4	Iris-setosa
32	33	5.2	4.1	1.5	0.1	Iris-setosa
33	34	5.5	4.2	1.4	0.2	Iris-setosa
34	35	4.9	3.1	1.5	0.1	Iris-setosa
35	36	5.0	3.2	1.2	0.2	Iris-setosa
36	37	5.5	3.5	1.3	0.2	Iris-setosa
37	38	4.9	3.1	1.5	0.1	Iris-setosa
38	39	4.4	3.0	1.3	0.2	Iris-setosa
39	40	5.1	3.4	1.5	0.2	Iris-setosa

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
40	41	5.0	3.5	1.3	0.3	Iris-setosa
41	42	4.5	2.3	1.3	0.3	Iris-setosa
42	43	4.4	3.2	1.3	0.2	Iris-setosa
43	44	5.0	3.5	1.6	0.6	Iris-setosa
44	45	5.1	3.8	1.9	0.4	Iris-setosa
45	46	4.8	3.0	1.4	0.3	Iris-setosa
46	47	5.1	3.8	1.6	0.2	Iris-setosa
47	48	4.6	3.2	1.4	0.2	Iris-setosa
48	49	5.3	3.7	1.5	0.2	Iris-setosa
49	50	5.0	3.3	1.4	0.2	Iris-setosa
50	51	7.0	3.2	4.7	1.4	Iris-versicolor
51	52	6.4	3.2	4.5	1.5	Iris-versicolor
52	53	6.9	3.1	4.9	1.5	Iris-versicolor
53	54	5.5	2.3	4.0	1.3	Iris-versicolor
54	55	6.5	2.8	4.6	1.5	Iris-versicolor
55	56	5.7	2.8	4.5	1.3	Iris-versicolor
56	57	6.3	3.3	4.7	1.6	Iris-versicolor
57	58	4.9	2.4	3.3	1.0	Iris-versicolor
58	59	6.6	2.9	4.6	1.3	Iris-versicolor
59	60	5.2	2.7	3.9	1.4	Iris-versicolor

```
In [5]: data_iris.tail(60)
```

Out[5]:

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
90	91	5.5	2.6	4.4	1.2	Iris-versicolor
91	92	6.1	3.0	4.6	1.4	Iris-versicolor
92	93	5.8	2.6	4.0	1.2	Iris-versicolor
93	94	5.0	2.3	3.3	1.0	Iris-versicolor
94	95	5.6	2.7	4.2	1.3	Iris-versicolor
95	96	5.7	3.0	4.2	1.2	Iris-versicolor
96	97	5.7	2.9	4.2	1.3	Iris-versicolor
97	98	6.2	2.9	4.3	1.3	Iris-versicolor
98	99	5.1	2.5	3.0	1.1	Iris-versicolor
99	100	5.7	2.8	4.1	1.3	Iris-versicolor
100	101	6.3	3.3	6.0	2.5	Iris-virginica
101	102	5.8	2.7	5.1	1.9	Iris-virginica
102	103	7.1	3.0	5.9	2.1	Iris-virginica
103	104	6.3	2.9	5.6	1.8	Iris-virginica
104	105	6.5	3.0	5.8	2.2	Iris-virginica
105	106	7.6	3.0	6.6	2.1	Iris-virginica
106	107	4.9	2.5	4.5	1.7	Iris-virginica
107	108	7.3	2.9	6.3	1.8	Iris-virginica
108	109	6.7	2.5	5.8	1.8	Iris-virginica
109	110	7.2	3.6	6.1	2.5	Iris-virginica
110	111	6.5	3.2	5.1	2.0	Iris-virginica
111	112	6.4	2.7	5.3	1.9	Iris-virginica
112	113	6.8	3.0	5.5	2.1	Iris-virginica
113	114	5.7	2.5	5.0	2.0	Iris-virginica
114	115	5.8	2.8	5.1	2.4	Iris-virginica
115	116	6.4	3.2	5.3	2.3	Iris-virginica
116	117	6.5	3.0	5.5	1.8	Iris-virginica
117	118	7.7	3.8	6.7	2.2	Iris-virginica
118	119	7.7	2.6	6.9	2.3	Iris-virginica
119	120	6.0	2.2	5.0	1.5	Iris-virginica
120	121	6.9	3.2	5.7	2.3	Iris-virginica
121	122	5.6	2.8	4.9	2.0	Iris-virginica
122	123	7.7	2.8	6.7	2.0	Iris-virginica
123	124	6.3	2.7	4.9	1.8	Iris-virginica
124	125	6.7	3.3	5.7	2.1	Iris-virginica
125	126	7.2	3.2	6.0	1.8	Iris-virginica
126	127	6.2	2.8	4.8	1.8	Iris-virginica
127	128	6.1	3.0	4.9	1.8	Iris-virginica
128	129	6.4	2.8	5.6	2.1	Iris-virginica
129	130	7.2	3.0	5.8	1.6	Iris-virginica

	Id	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	Species
130	131	7.4	2.8	6.1	1.9	Iris-virginica
131	132	7.9	3.8	6.4	2.0	Iris-virginica
132	133	6.4	2.8	5.6	2.2	Iris-virginica
133	134	6.3	2.8	5.1	1.5	Iris-virginica
134	135	6.1	2.6	5.6	1.4	Iris-virginica
135	136	7.7	3.0	6.1	2.3	Iris-virginica
136	137	6.3	3.4	5.6	2.4	Iris-virginica
137	138	6.4	3.1	5.5	1.8	Iris-virginica
138	139	6.0	3.0	4.8	1.8	Iris-virginica
139	140	6.9	3.1	5.4	2.1	Iris-virginica
140	141	6.7	3.1	5.6	2.4	Iris-virginica
141	142	6.9	3.1	5.1	2.3	Iris-virginica
142	143	5.8	2.7	5.1	1.9	Iris-virginica
143	144	6.8	3.2	5.9	2.3	Iris-virginica
144	145	6.7	3.3	5.7	2.5	Iris-virginica
145	146	6.7	3.0	5.2	2.3	Iris-virginica
146	147	6.3	2.5	5.0	1.9	Iris-virginica
147	148	6.5	3.0	5.2	2.0	Iris-virginica
148	149	6.2	3.4	5.4	2.3	Iris-virginica
149	150	5.9	3.0	5.1	1.8	Iris-virginica

In [5]: data_iris.info()

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 6 columns):
#   Column          Non-Null Count  Dtype
---  ---
0    Id              150 non-null    int64
1   SepalLengthCm   150 non-null    float64
2   SepalWidthCm    150 non-null    float64
3   PetalLengthCm   150 non-null    float64
4   PetalWidthCm    150 non-null    float64
5   Species         150 non-null    object
dtypes: float64(4), int64(1), object(1)
memory usage: 7.2+ KB
```

In [4]: data_iris['Species'].unique()

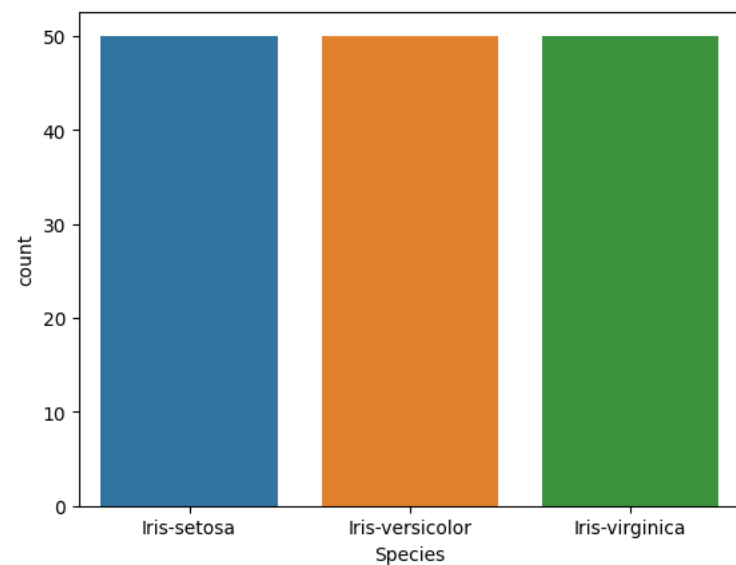
Out[4]: array(['Iris-setosa', 'Iris-versicolor', 'Iris-virginica'], dtype=object)

In [5]: data_iris['Species'].value_counts()

Out[5]: Iris-setosa 50
Iris-versicolor 50
Iris-virginica 50
Name: Species, dtype: int64

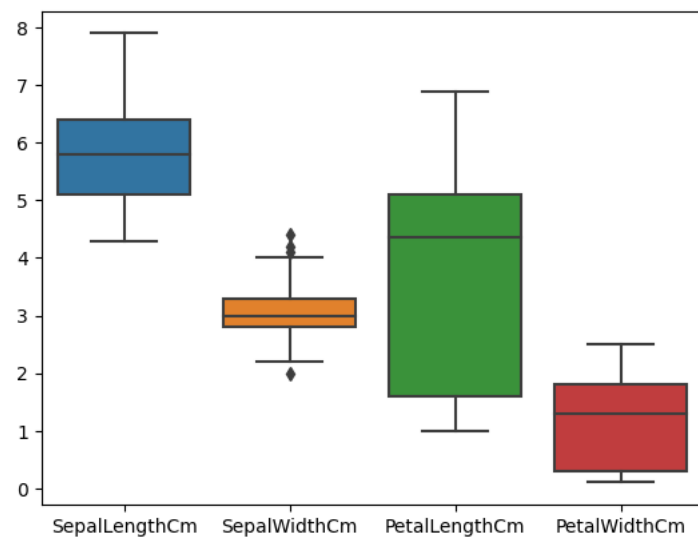
```
In [6]: sns.countplot(x = data_iris['Species'])
```

```
Out[6]: <AxesSubplot: xlabel='Species', ylabel='count'>
```



```
In [6]: sns.boxplot(data_iris.iloc[:,1:5])
```

```
Out[6]: <AxesSubplot: >
```



Making Dataset for Clustering

```
In [6]: iris = data_iris.iloc[:,1:5] # removing the final lable or result lable
iris
```

```
Out[6]:
```

	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm
0	5.1	3.5	1.4	0.2
1	4.9	3.0	1.4	0.2
2	4.7	3.2	1.3	0.2
3	4.6	3.1	1.5	0.2
4	5.0	3.6	1.4	0.2
...	...	...	...	...
145	6.7	3.0	5.2	2.3
146	6.3	2.5	5.0	1.9
147	6.5	3.0	5.2	2.0
148	6.2	3.4	5.4	2.3
149	5.9	3.0	5.1	1.8

150 rows × 4 columns

importing the K-Means

```
In [7]: from sklearn.cluster import KMeans
```

fit the data in algo

```
In [8]: iris_cluster = KMeans(n_clusters=10)
```

```
In [10]: iris_cluster.fit(iris)
```

```
Out[10]:
```

KMeans

KMeans(n_clusters=10)

```
In [11]: iris_cluster.labels_
```

```
Out[11]: array([1, 8, 8, 8, 1, 9, 1, 1, 8, 1, 9, 1, 8, 8, 9, 9, 9, 1, 9, 9, 9, 9,
 1, 1, 1, 1, 1, 1, 1, 8, 8, 9, 9, 9, 1, 1, 9, 1, 8, 1, 1, 8, 8, 1,
 9, 8, 9, 8, 9, 1, 0, 0, 0, 7, 0, 2, 0, 7, 0, 7, 7, 2, 7, 0, 7, 0,
 2, 2, 0, 7, 5, 2, 5, 0, 0, 0, 0, 0, 2, 7, 7, 7, 2, 5, 2, 0, 0, 0,
 2, 7, 2, 0, 2, 7, 2, 2, 2, 0, 7, 2, 4, 5, 3, 4, 4, 3, 2, 3, 4, 6,
 4, 5, 4, 5, 5, 4, 4, 6, 3, 5, 4, 5, 3, 5, 4, 3, 5, 5, 4, 3, 3, 6,
 4, 5, 5, 3, 4, 4, 5, 4, 4, 4, 5, 4, 4, 4, 5, 4, 4, 5])
```

```
In [12]: c_10 = pd.DataFrame(iris_cluster.labels_)
```

```
In [13]: c_10
```

```
Out[13]:
```

	0
0	1
1	8
2	8
3	8
4	1
...	...
145	4
146	5
147	4
148	4
149	5

150 rows × 1 columns

```
In [14]: iris_change = pd.concat([iris,c_10],axis = 1)
```

```
In [15]: iris_change = iris_change.rename(columns = {0: 'C_10'})
```

```
In [17]: iris_change.head(60)
```

Out[17]:

	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	C_10
0	5.1	3.5	1.4	0.2	1
1	4.9	3.0	1.4	0.2	8
2	4.7	3.2	1.3	0.2	8
3	4.6	3.1	1.5	0.2	8
4	5.0	3.6	1.4	0.2	1
5	5.4	3.9	1.7	0.4	9
6	4.6	3.4	1.4	0.3	1
7	5.0	3.4	1.5	0.2	1
8	4.4	2.9	1.4	0.2	8
9	4.9	3.1	1.5	0.1	1
10	5.4	3.7	1.5	0.2	9
11	4.8	3.4	1.6	0.2	1
12	4.8	3.0	1.4	0.1	8
13	4.3	3.0	1.1	0.1	8
14	5.8	4.0	1.2	0.2	9
15	5.7	4.4	1.5	0.4	9
16	5.4	3.9	1.3	0.4	9
17	5.1	3.5	1.4	0.3	1
18	5.7	3.8	1.7	0.3	9
19	5.1	3.8	1.5	0.3	9
20	5.4	3.4	1.7	0.2	9
21	5.1	3.7	1.5	0.4	9
22	4.6	3.6	1.0	0.2	1
23	5.1	3.3	1.7	0.5	1
24	4.8	3.4	1.9	0.2	1
25	5.0	3.0	1.6	0.2	1
26	5.0	3.4	1.6	0.4	1
27	5.2	3.5	1.5	0.2	1
28	5.2	3.4	1.4	0.2	1
29	4.7	3.2	1.6	0.2	8
30	4.8	3.1	1.6	0.2	8
31	5.4	3.4	1.5	0.4	9
32	5.2	4.1	1.5	0.1	9
33	5.5	4.2	1.4	0.2	9
34	4.9	3.1	1.5	0.1	1
35	5.0	3.2	1.2	0.2	1
36	5.5	3.5	1.3	0.2	9
37	4.9	3.1	1.5	0.1	1
38	4.4	3.0	1.3	0.2	8
39	5.1	3.4	1.5	0.2	1

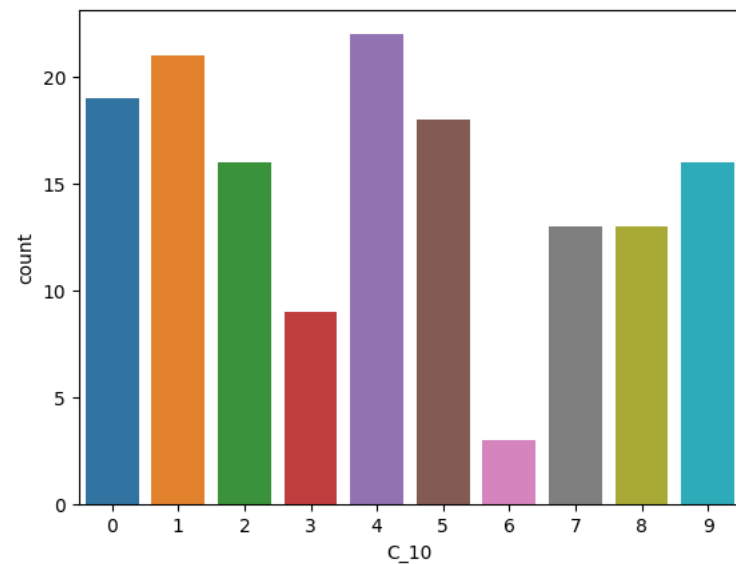
	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	C_10
40	5.0	3.5	1.3	0.3	1
41	4.5	2.3	1.3	0.3	8
42	4.4	3.2	1.3	0.2	8
43	5.0	3.5	1.6	0.6	1
44	5.1	3.8	1.9	0.4	9
45	4.8	3.0	1.4	0.3	8
46	5.1	3.8	1.6	0.2	9
47	4.6	3.2	1.4	0.2	8
48	5.3	3.7	1.5	0.2	9
49	5.0	3.3	1.4	0.2	1
50	7.0	3.2	4.7	1.4	0
51	6.4	3.2	4.5	1.5	0
52	6.9	3.1	4.9	1.5	0
53	5.5	2.3	4.0	1.3	7
54	6.5	2.8	4.6	1.5	0
55	5.7	2.8	4.5	1.3	2
56	6.3	3.3	4.7	1.6	0
57	4.9	2.4	3.3	1.0	7
58	6.6	2.9	4.6	1.3	0
59	5.2	2.7	3.9	1.4	7

```
In [19]: iris_change['C_10'].value_counts()
```

```
Out[19]: 4    22
         1    21
         0    19
         5    18
         9    16
         2    16
         8    13
         7    13
         3     9
         6     3
         Name: C_10, dtype: int64
```

```
In [18]: sns.countplot(x = iris_change['C_10'])
```

```
Out[18]: <AxesSubplot: xlabel='C_10', ylabel='count'>
```



```
In [ ]:
```

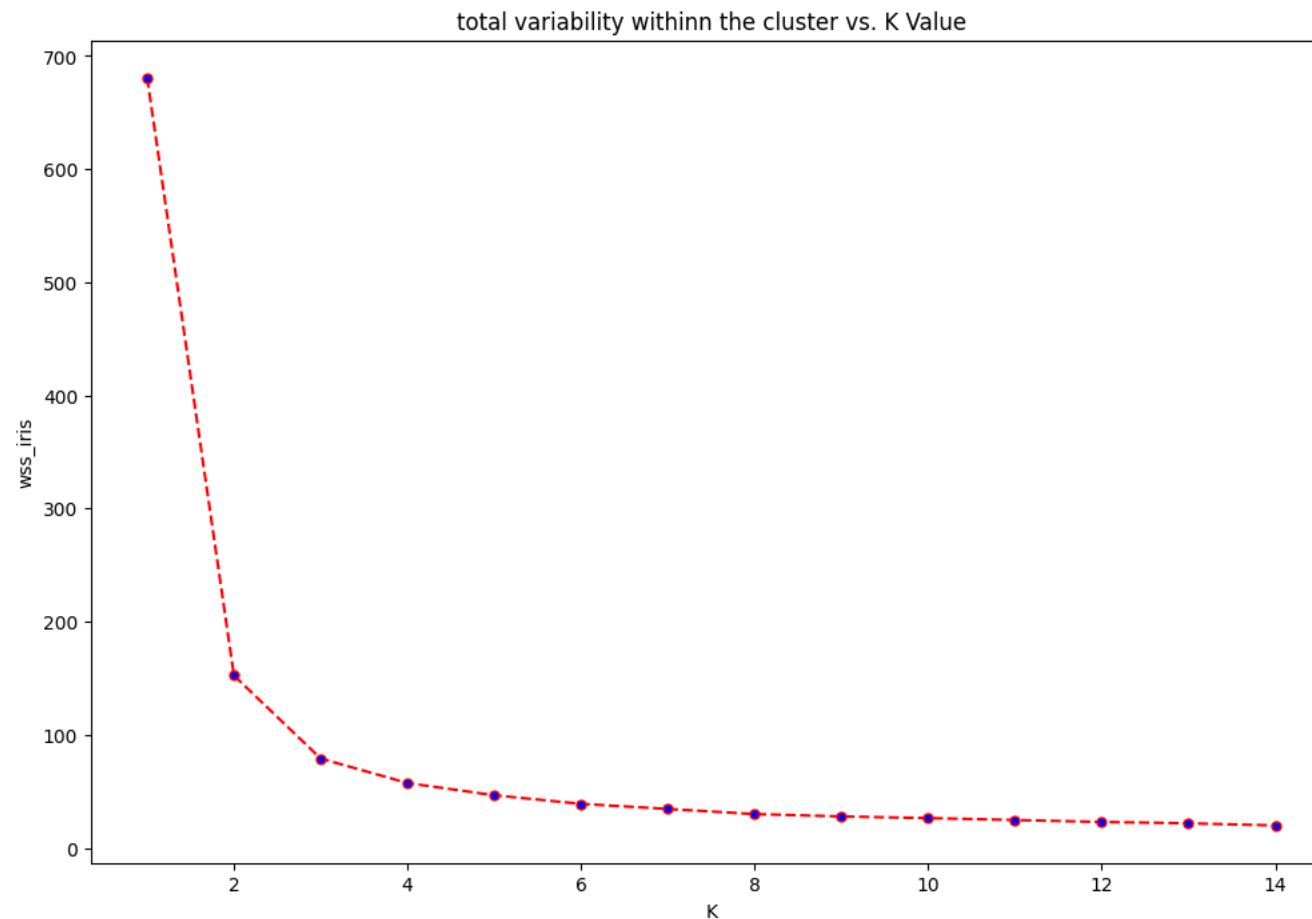
finding the best value of number of clusters by elbow method based on WCSS (Within Cluster Sum of Squares)

```
In [20]: # finding the best no of cluster
wss_iris=[]

for i in range(1,15):
    k_iris_c = KMeans(n_clusters=i)
    k_iris_c.fit(iris)
    pred_i = k_iris_c.labels_
    wss_iris.append(k_iris_c.inertia_)
```

```
In [21]: plt.figure(figsize=(12,8))
plt.plot(range(1,15),wss_iris,color='red', linestyle='dashed', marker='o',
         markerfacecolor='blue', markersize=5)
plt.title('total variability withinn the cluster vs. K Value')
plt.xlabel('K')
plt.ylabel('wss_iris')
```

Out[21]: Text(0, 0.5, 'wss_iris')



Again using the KMeas for best number of Cluster

```
In [22]: iris_best_n = KMeans(n_clusters=3)
```

```
In [23]: iris_best_n.fit(iris)
```

Out[23]:

KMeans

KMeans(n_clusters=3)

In [24]: iris_best_n.labels_

Out[24]: array([1,
1,
1, 1, 1, 1, 1, 1, 0, 0, 2, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 2, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0,
0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 2, 0, 2, 2, 2, 2, 0, 2, 2, 2, 2,
2, 2, 2, 0, 0, 2, 2, 2, 2, 0, 2, 0, 2, 2, 2, 0, 0, 2, 2, 2, 2, 2,
2, 0, 2, 2, 2, 2, 0, 2, 2, 2, 0, 2, 2, 2, 0, 2, 2, 0])

In [25]: c_3 = pd.DataFrame(iris_best_n.labels_)

In [26]: c_3 = c_3.rename(columns = {0:'C_3'})
c_3

Out[26]:

	C_3
0	1
1	1
2	1
3	1
4	1
...	...
145	2
146	0
147	2
148	2
149	0

150 rows × 1 columns

In [27]: iris_change = pd.concat([iris_change, c_3],axis = 1)

In [28]: iris_change

Out[28]:

	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	C_10	C_3
0	5.1	3.5	1.4	0.2	1	1
1	4.9	3.0	1.4	0.2	8	1
2	4.7	3.2	1.3	0.2	8	1
3	4.6	3.1	1.5	0.2	8	1
4	5.0	3.6	1.4	0.2	1	1
...	...	...	...	...	...	...
145	6.7	3.0	5.2	2.3	4	2
146	6.3	2.5	5.0	1.9	5	0
147	6.5	3.0	5.2	2.0	4	2
148	6.2	3.4	5.4	2.3	4	2
149	5.9	3.0	5.1	1.8	5	0

150 rows × 6 columns

```
In [29]: iris_change['C_3'].value_counts()
```

```
Out[29]: 0    62  
         1    50  
         2    38  
         Name: C_3, dtype: int64
```

```
In [30]: iris_change['Actual_Species'] = data_iris['Species']
```

```
In [32]: iris_change.head(60)
```

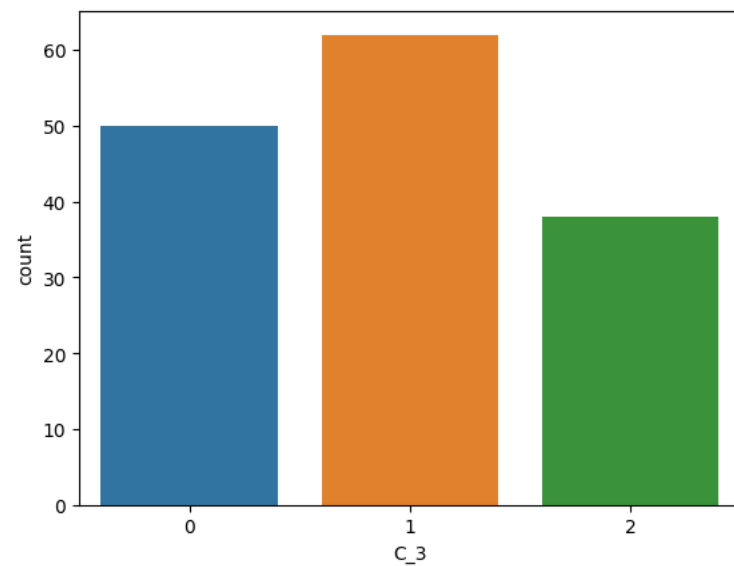
Out[32]:

	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	C_10	C_3	Actual_Species
0	5.1	3.5	1.4	0.2	1	1	Iris-setosa
1	4.9	3.0	1.4	0.2	8	1	Iris-setosa
2	4.7	3.2	1.3	0.2	8	1	Iris-setosa
3	4.6	3.1	1.5	0.2	8	1	Iris-setosa
4	5.0	3.6	1.4	0.2	1	1	Iris-setosa
5	5.4	3.9	1.7	0.4	9	1	Iris-setosa
6	4.6	3.4	1.4	0.3	1	1	Iris-setosa
7	5.0	3.4	1.5	0.2	1	1	Iris-setosa
8	4.4	2.9	1.4	0.2	8	1	Iris-setosa
9	4.9	3.1	1.5	0.1	1	1	Iris-setosa
10	5.4	3.7	1.5	0.2	9	1	Iris-setosa
11	4.8	3.4	1.6	0.2	1	1	Iris-setosa
12	4.8	3.0	1.4	0.1	8	1	Iris-setosa
13	4.3	3.0	1.1	0.1	8	1	Iris-setosa
14	5.8	4.0	1.2	0.2	9	1	Iris-setosa
15	5.7	4.4	1.5	0.4	9	1	Iris-setosa
16	5.4	3.9	1.3	0.4	9	1	Iris-setosa
17	5.1	3.5	1.4	0.3	1	1	Iris-setosa
18	5.7	3.8	1.7	0.3	9	1	Iris-setosa
19	5.1	3.8	1.5	0.3	9	1	Iris-setosa
20	5.4	3.4	1.7	0.2	9	1	Iris-setosa
21	5.1	3.7	1.5	0.4	9	1	Iris-setosa
22	4.6	3.6	1.0	0.2	1	1	Iris-setosa
23	5.1	3.3	1.7	0.5	1	1	Iris-setosa
24	4.8	3.4	1.9	0.2	1	1	Iris-setosa
25	5.0	3.0	1.6	0.2	1	1	Iris-setosa
26	5.0	3.4	1.6	0.4	1	1	Iris-setosa
27	5.2	3.5	1.5	0.2	1	1	Iris-setosa
28	5.2	3.4	1.4	0.2	1	1	Iris-setosa
29	4.7	3.2	1.6	0.2	8	1	Iris-setosa
30	4.8	3.1	1.6	0.2	8	1	Iris-setosa
31	5.4	3.4	1.5	0.4	9	1	Iris-setosa
32	5.2	4.1	1.5	0.1	9	1	Iris-setosa
33	5.5	4.2	1.4	0.2	9	1	Iris-setosa
34	4.9	3.1	1.5	0.1	1	1	Iris-setosa
35	5.0	3.2	1.2	0.2	1	1	Iris-setosa
36	5.5	3.5	1.3	0.2	9	1	Iris-setosa
37	4.9	3.1	1.5	0.1	1	1	Iris-setosa
38	4.4	3.0	1.3	0.2	8	1	Iris-setosa
39	5.1	3.4	1.5	0.2	1	1	Iris-setosa

	SepalLengthCm	SepalWidthCm	PetalLengthCm	PetalWidthCm	C_10	C_3	Actual_Species
40	5.0	3.5	1.3	0.3	1	1	Iris-setosa
41	4.5	2.3	1.3	0.3	8	1	Iris-setosa
42	4.4	3.2	1.3	0.2	8	1	Iris-setosa
43	5.0	3.5	1.6	0.6	1	1	Iris-setosa
44	5.1	3.8	1.9	0.4	9	1	Iris-setosa
45	4.8	3.0	1.4	0.3	8	1	Iris-setosa
46	5.1	3.8	1.6	0.2	9	1	Iris-setosa
47	4.6	3.2	1.4	0.2	8	1	Iris-setosa
48	5.3	3.7	1.5	0.2	9	1	Iris-setosa
49	5.0	3.3	1.4	0.2	1	1	Iris-setosa
50	7.0	3.2	4.7	1.4	0	0	Iris-versicolor
51	6.4	3.2	4.5	1.5	0	0	Iris-versicolor
52	6.9	3.1	4.9	1.5	0	2	Iris-versicolor
53	5.5	2.3	4.0	1.3	7	0	Iris-versicolor
54	6.5	2.8	4.6	1.5	0	0	Iris-versicolor
55	5.7	2.8	4.5	1.3	2	0	Iris-versicolor
56	6.3	3.3	4.7	1.6	0	0	Iris-versicolor
57	4.9	2.4	3.3	1.0	7	0	Iris-versicolor
58	6.6	2.9	4.6	1.3	0	0	Iris-versicolor
59	5.2	2.7	3.9	1.4	7	0	Iris-versicolor

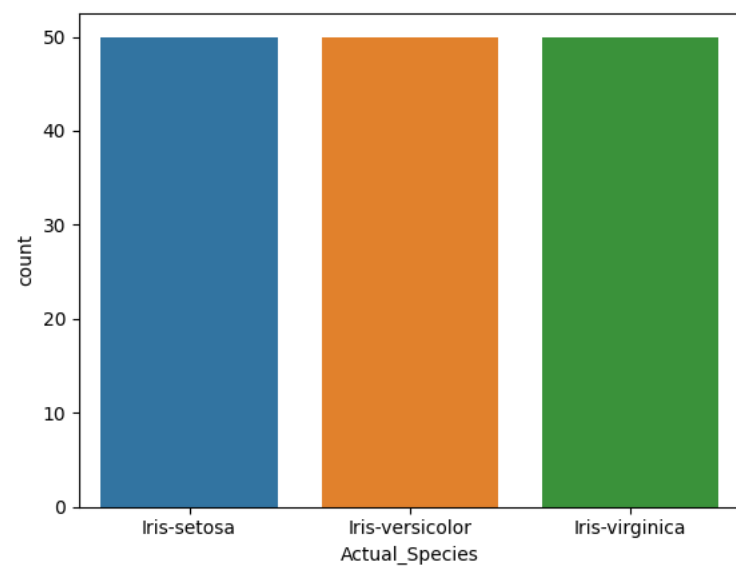
```
In [32]: sns.countplot(x = iris_change['C_3'])
```

```
Out[32]: <AxesSubplot: xlabel='C_3', ylabel='count'>
```



```
In [34]: sns.countplot(x = iris_change['Actual_Species'])
```

```
Out[34]: <AxesSubplot: xlabel='Actual_Species', ylabel='count'>
```



You should note, the colors are meaningless in reference between the two plots.

Great Job!

What is 'K-Means' ? – Choosing number of Clusters

k-means clustering aims to partition ' n ' observations into ' k ' clusters.

Note : the value of ' k ' is determined by the analyst

How to choose optimum no. of Clusters? – Within group Sum of Squares (WSS) Plot

